Pit Analyzer

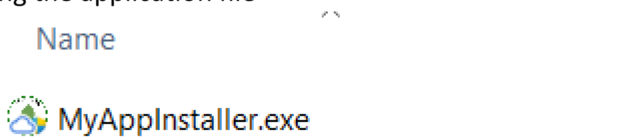
User manual

Last updated: July 12th 2026

1. Installation

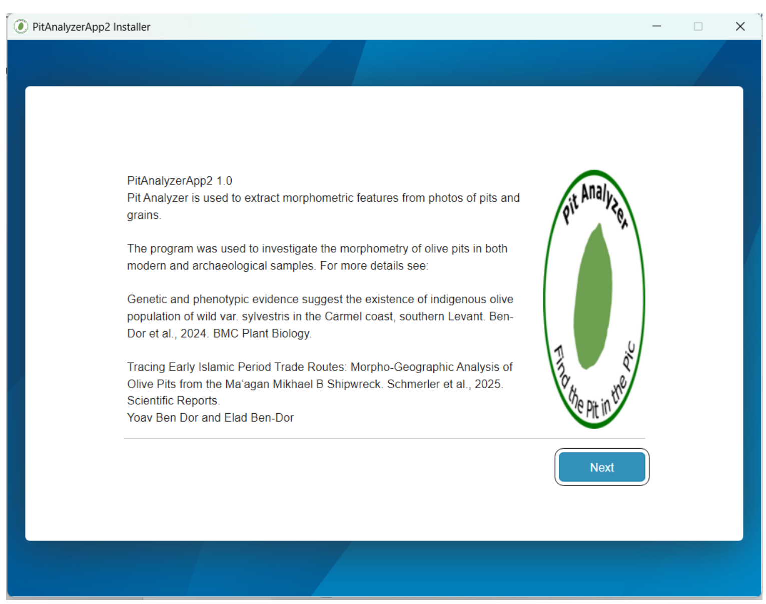
a) Download the installation file from the “package” folder in the repository

b) Install the app by clicking the application file



c) The installer, based on the MATLAB compiler, will open

d) Select the required installation method and destination folder and wait until all installations are done.



e) The installer will install the MATLAB runtime environment if required (the current version is based on 2025b)

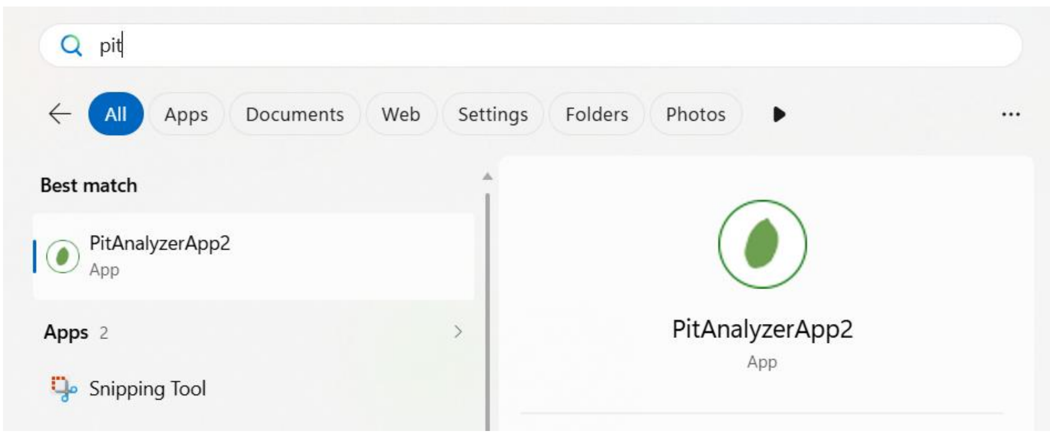
f) Tip: in case of uninstalling, there is no need to uninstall the MATLAB runtime, which can be used to run other MATLAB based applications

g) Tip: Future versions may rely on more advanced MATLAB versions, so the installer will update the MATLAB runtime as necessary

h) Tip: for updates and comments, please visit the GitHUB repository at: https://github.com/yoavbendor1/PitAnalyzer

2. Startup

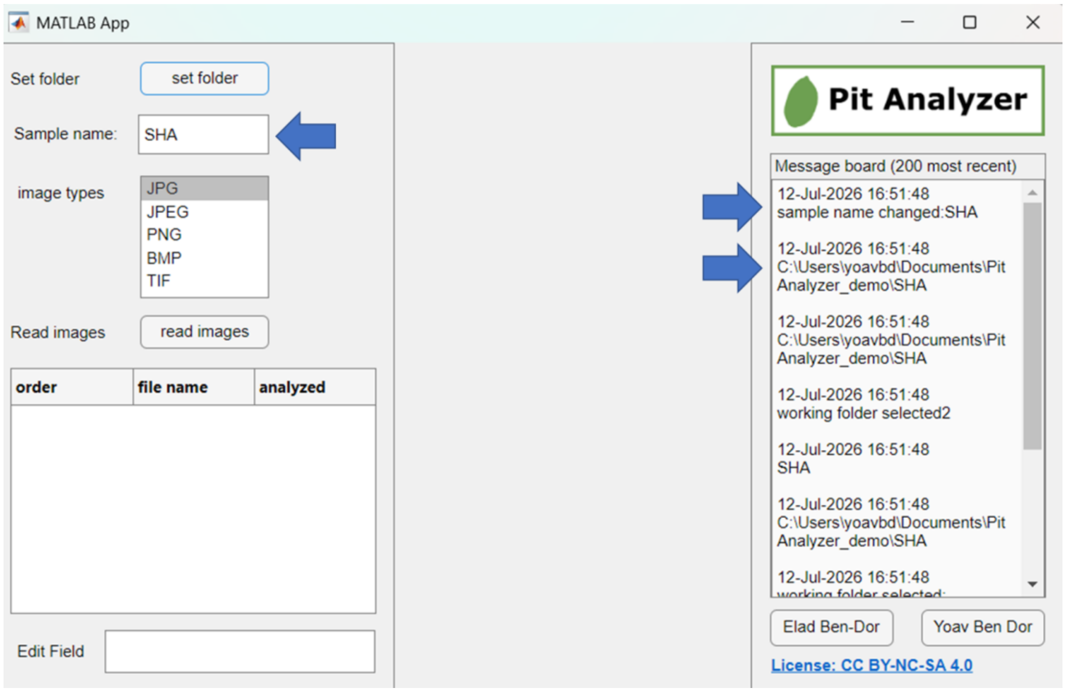
a) Open the program from the applications menu of the operating system



3. Set working directory

a) Set the working directory containing the photos to be analyzed by clicking the “set folder” button and selecting the folder that contains the image files for analyses

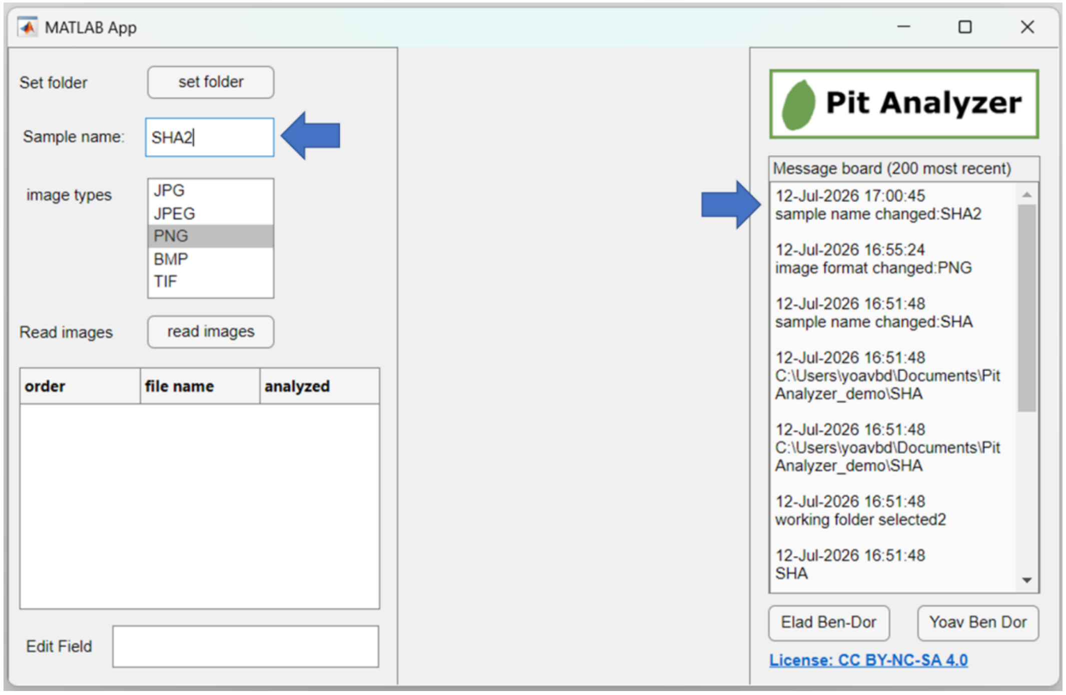
b) The updated folder name will be declared on the message board, and the sample name will be updated accordingly



c) Tip: best practice is to name each folder with the sample name
d) Tip: best practice is to maintain a consistent labelling of the image and removing any dust or dirt affecting the shape of the investigated pit from the photos. This can be done using a simple photo editing software such as paint or GIMP

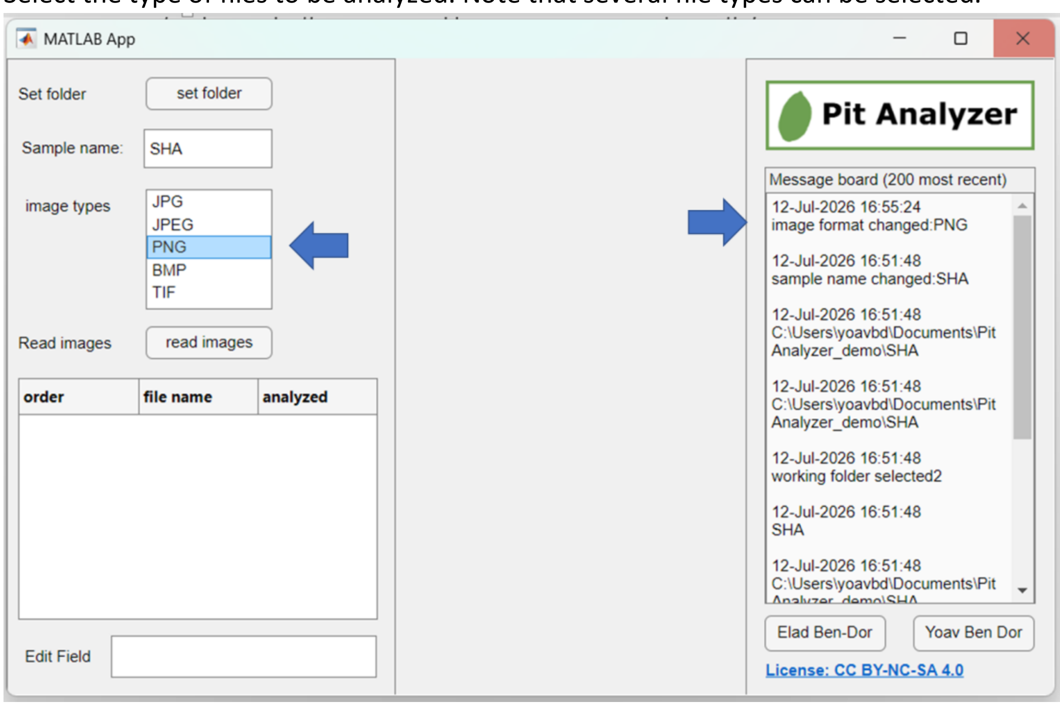
4. Adjust sample

a) If necessary, adjust sample name using the dedicated text box by typing its name and pressing the enter key
b) Sample name can be changed several times prior to image processing, and will affect the labelling of all files resulting from the analyses process



5. Set file type(s) for analyses

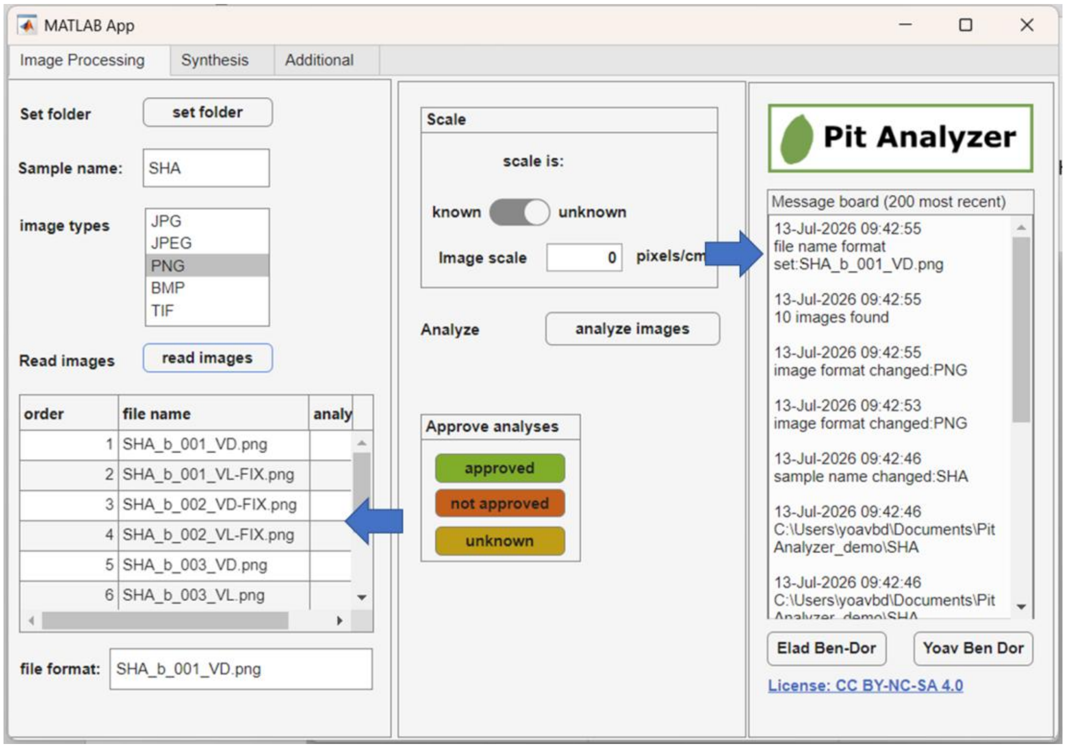
a) Select the type of files to be analyzed. Note that several file types can be selected.



6. Load available image files for analyses

a) Click the “read images” button to look for all files matching the selected type(s)

b) The program will now scan the selected folder and provide a list of the files identified for the user to examine
c) The number of identified images for analyses will be declared in the messages window



7. Update the file name format

a) This section will adjust the way the program "interprets" the file names in order to properly label the analyzed images.
b) In the presented example the photos are labelled using the following logic, with underscore as separator of label parts:

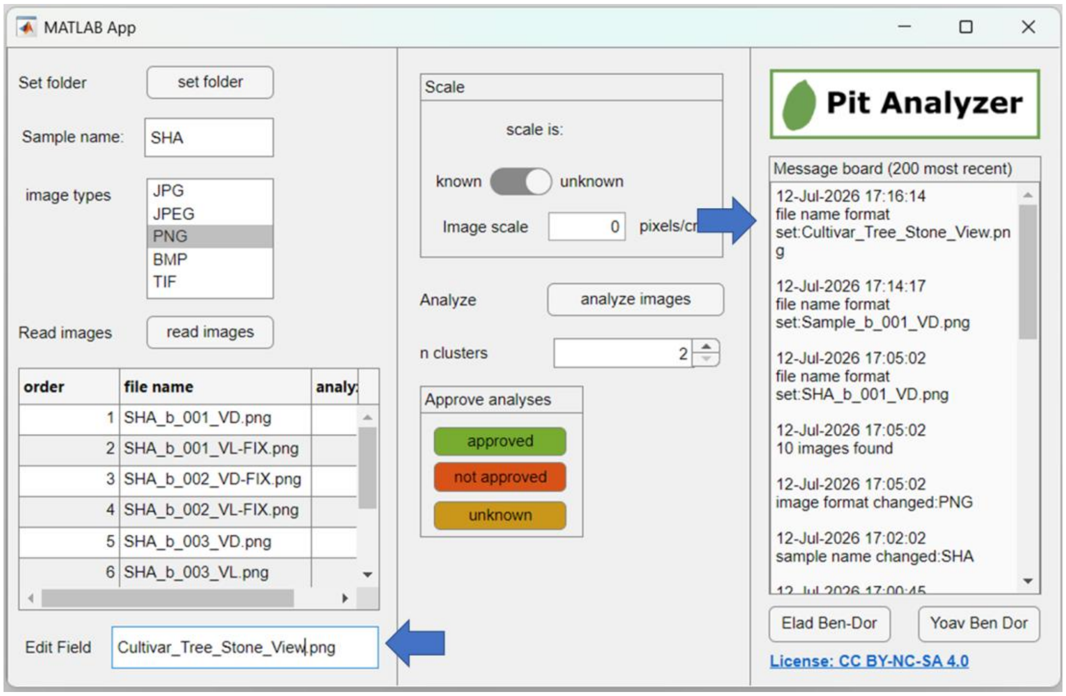
Label	Meaning	Name format field	Comments
SHA	Sample name	Cultivar	In this case also the code of the *Olea Europaea* cultivar *Shami* (also known as *Aggizi Shami* or *Chami*)
b	Tree code	Tree	The code of the tree from which the specimens were collected from
001	Fruit\pit code	Pit	The unique value of the specimen, as labelled in the lab. i.e., fruit\pit specimen 1.

VD	View direction of the pit	View	Marks the ventral\dorsal angle of the photo

c) In the abovementioned example the user should type the text: Cultivar_Tree_Pit_View in the "file format" window

d) Tip: make sure one of the columns contains the word "View" in order to allow the program to organize the tables properly

e) Tip: use underscore to separate the segments of the photo labels, and use a consistent and clear labelling approach to your photos in order to avoid mistakes and incorrect labelling



8. Set the scale of the analyzed images

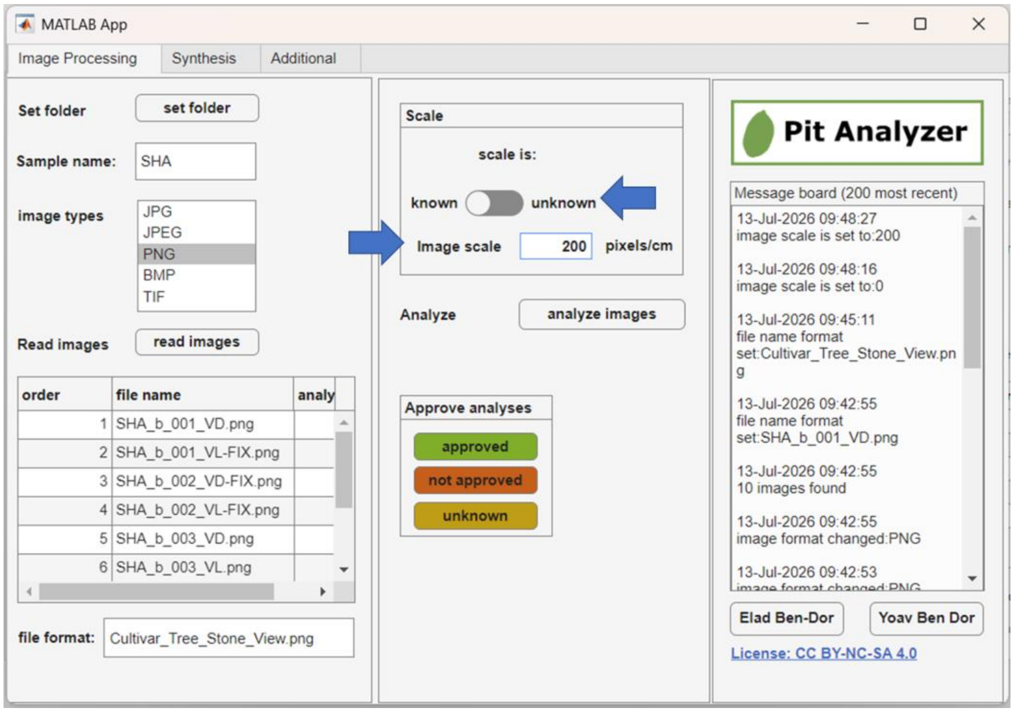
a) The scale is used to calculate the measured size and related features (e.g., radius, height, etc.) in terms of cm

b) Note that the images only provide a pixel scale, which makes it difficult to compare the results with previous publications. The scale is therefore crucial for providing the actual measured values in meters (or cm)

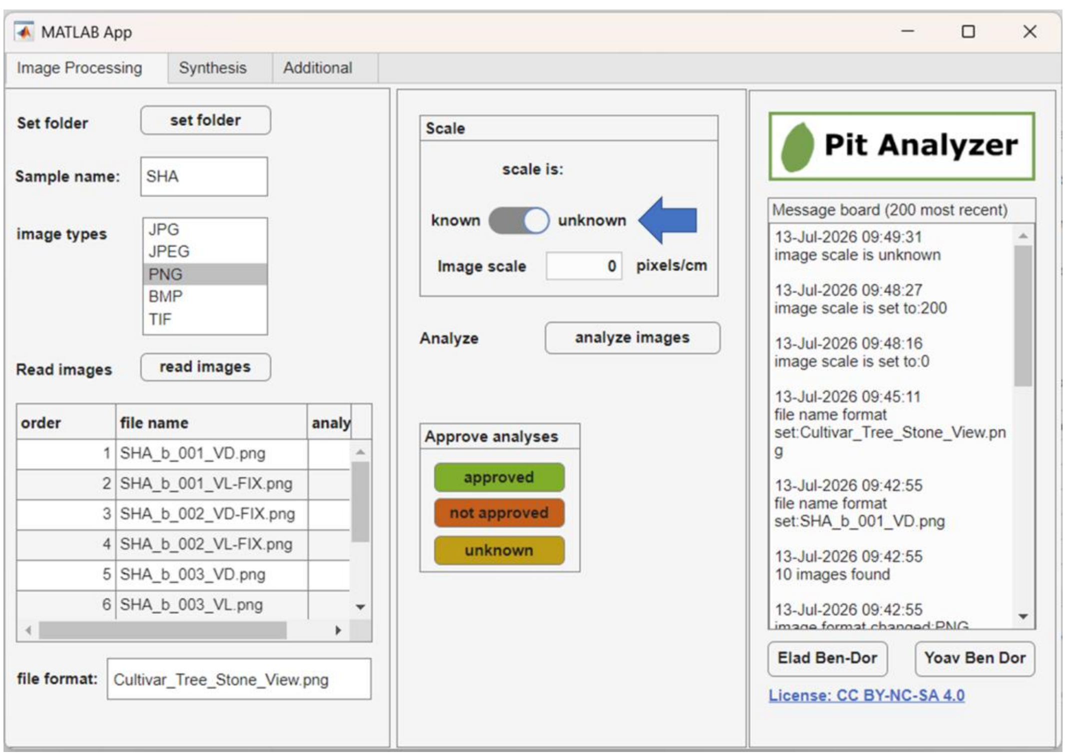
c) In case that the scale is known, one can toggle the "scale is known" button by clicking it, and then providing the program with the number of pixels in each cm of the photo.

d) Note that this should be done only if the scale was properly measured previously either within PitAnalyzer or using another photo editing program.

e) The program will notify the user on the newly established scale



f) In the case that the scale of the images is not known, the user should toggle the button to "unknown" and will be asked to measure a marker of known length of one cm in order to set the image scale

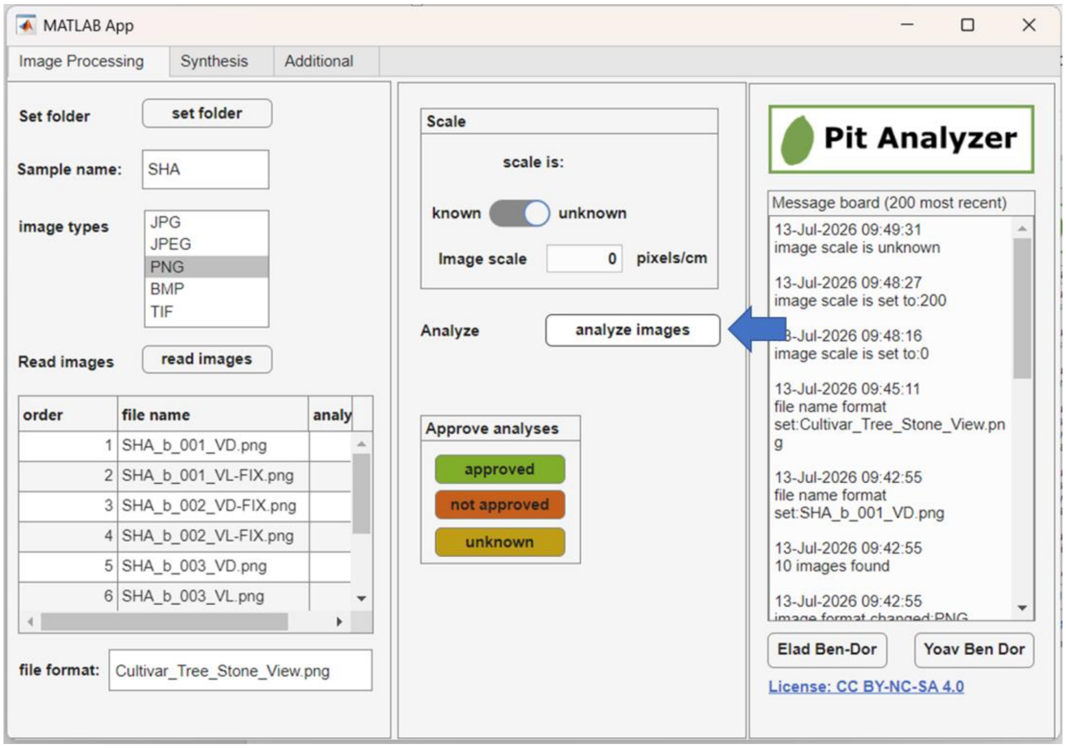


9. Image analyses

a) Once the scale is set (or set as unknown), the user can begin to analyze the data by clicking the "analyze images" button

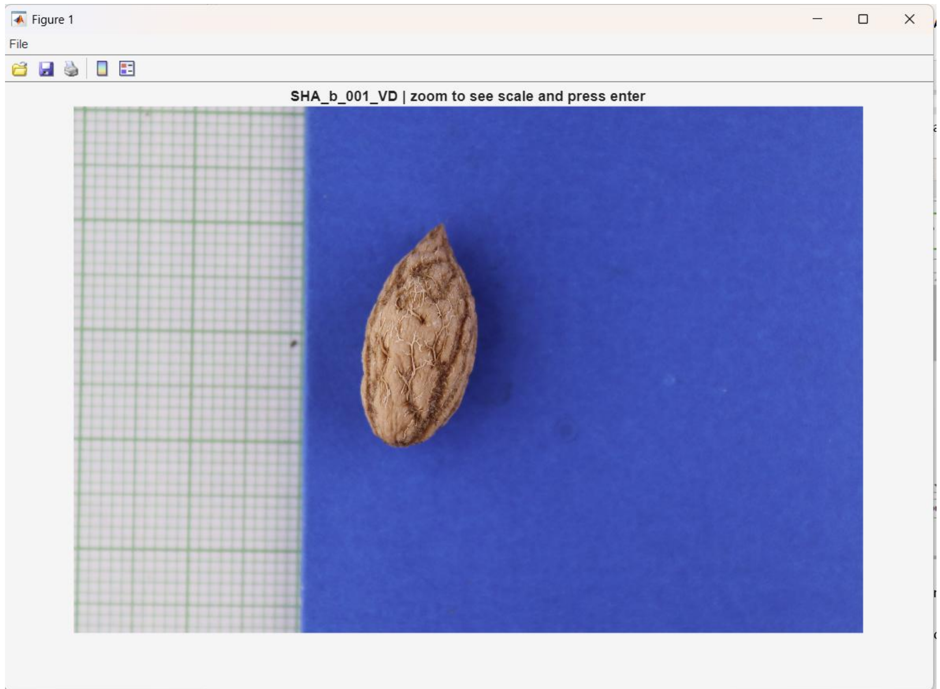
b) Providing that the contrast between the pit and the background is sufficient, the value of "n clusters" should be set to 2 (as the default value)

c) The program will look for previous csv files in order to avoid repeating the analyses of images that were already successfully analyzed

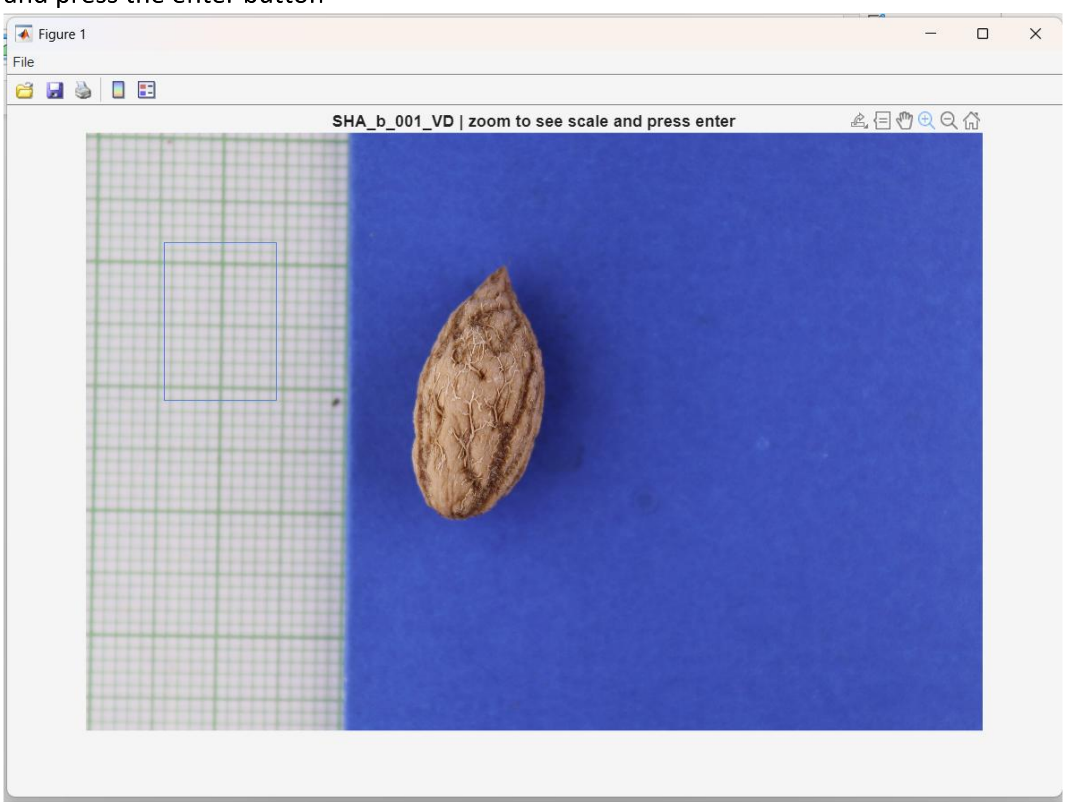


d) The program will now start loading the images found in the directory

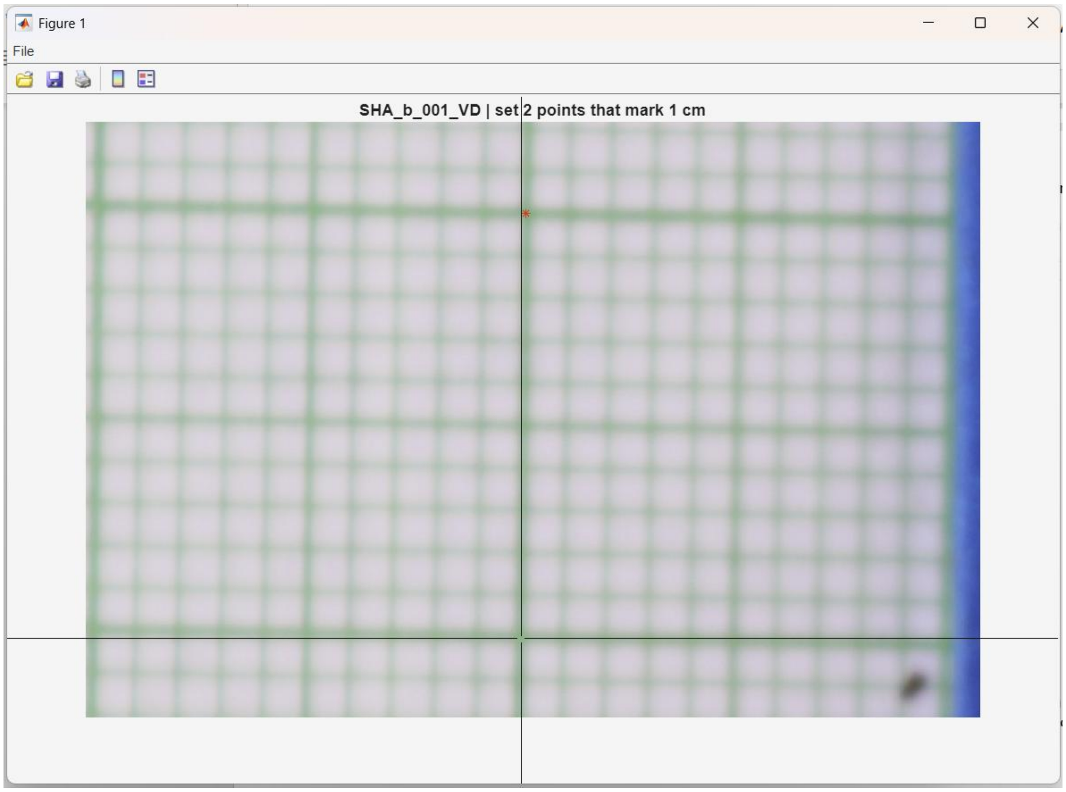
e) In the case that no scale was provided, a scale should be measured individually for each image. The median of these scales will then be used the representative scale for this folder.



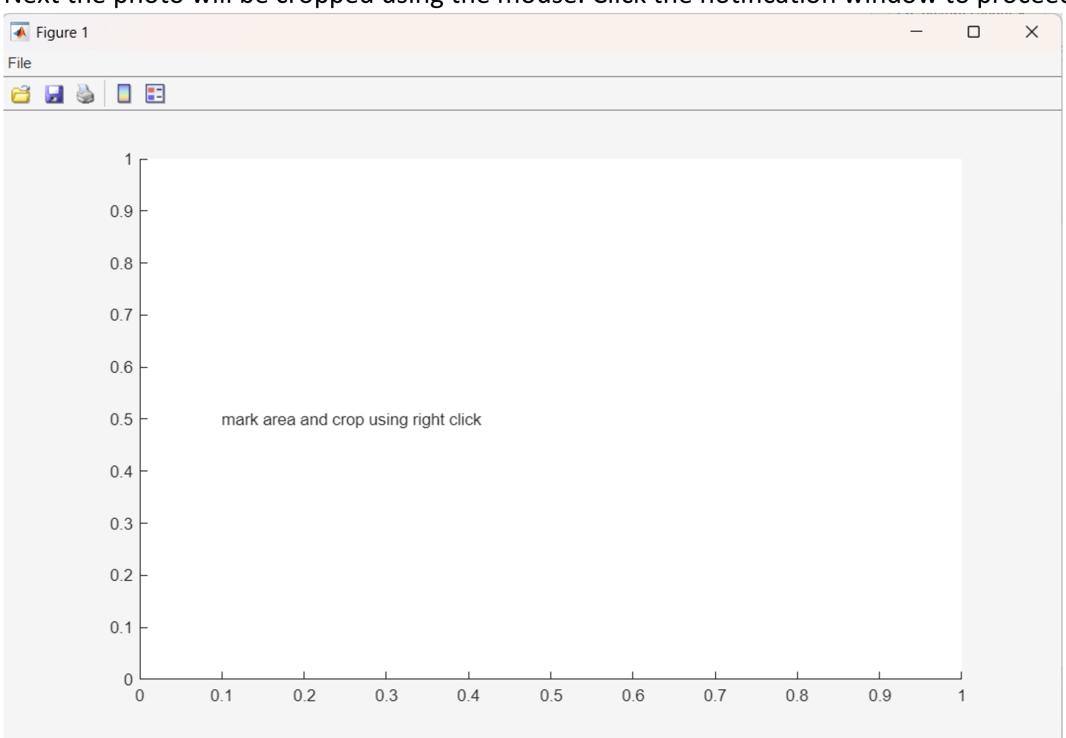
f) If the scale should be measured, zoom into an item of a fixed length of one cm using the mouse and press the enter button



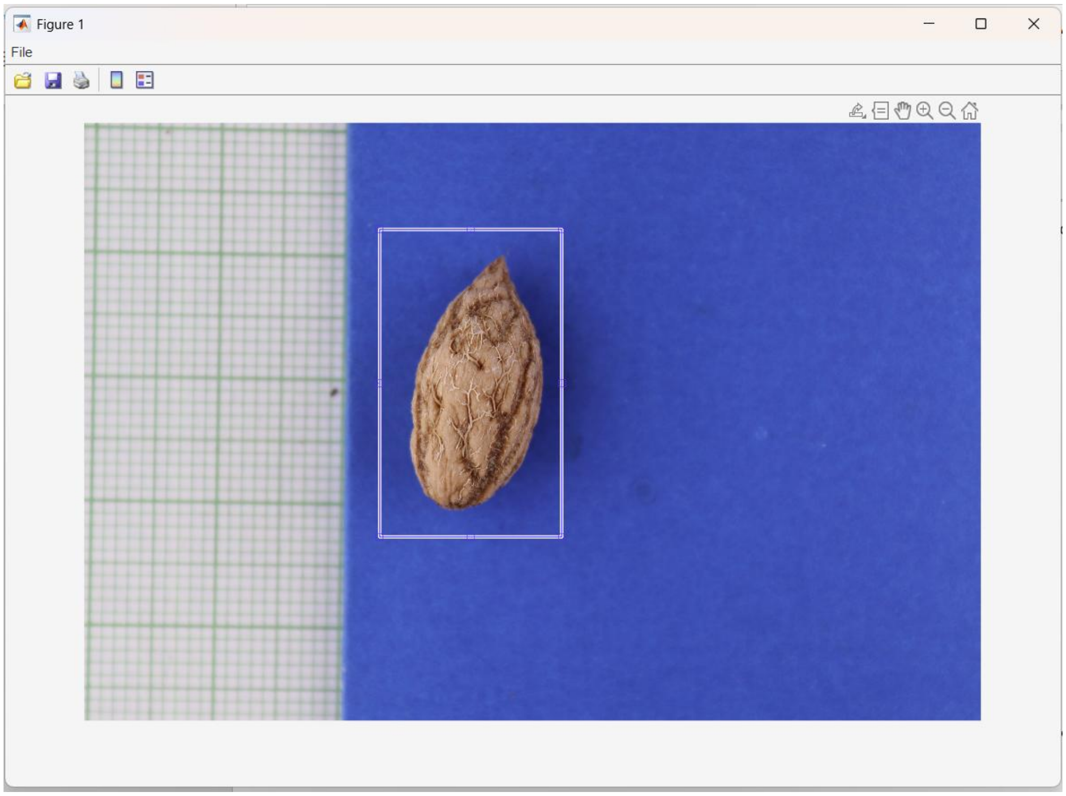
g) Use the crosshairs to mark a distance of one cm on the image. The program will calculate the Euclidean distance of the two points, so don't worry if they are not horizontal\vertical



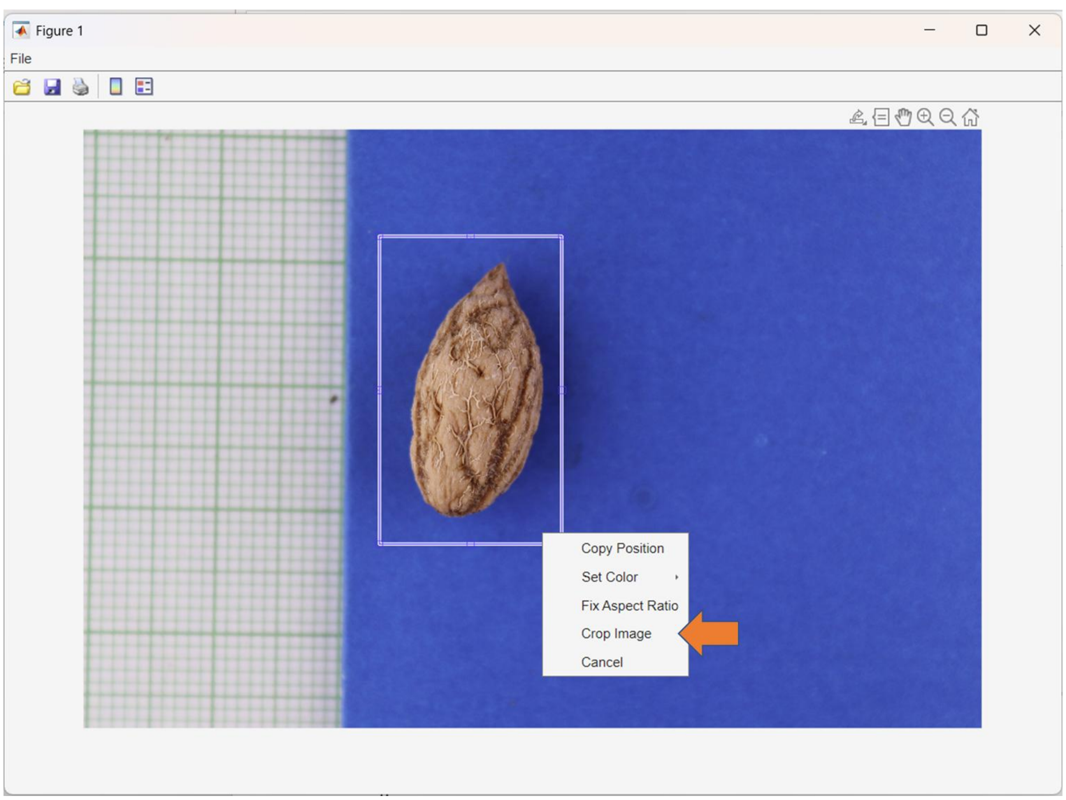
h) Next the photo will be cropped using the mouse. Click the notification window to proceed



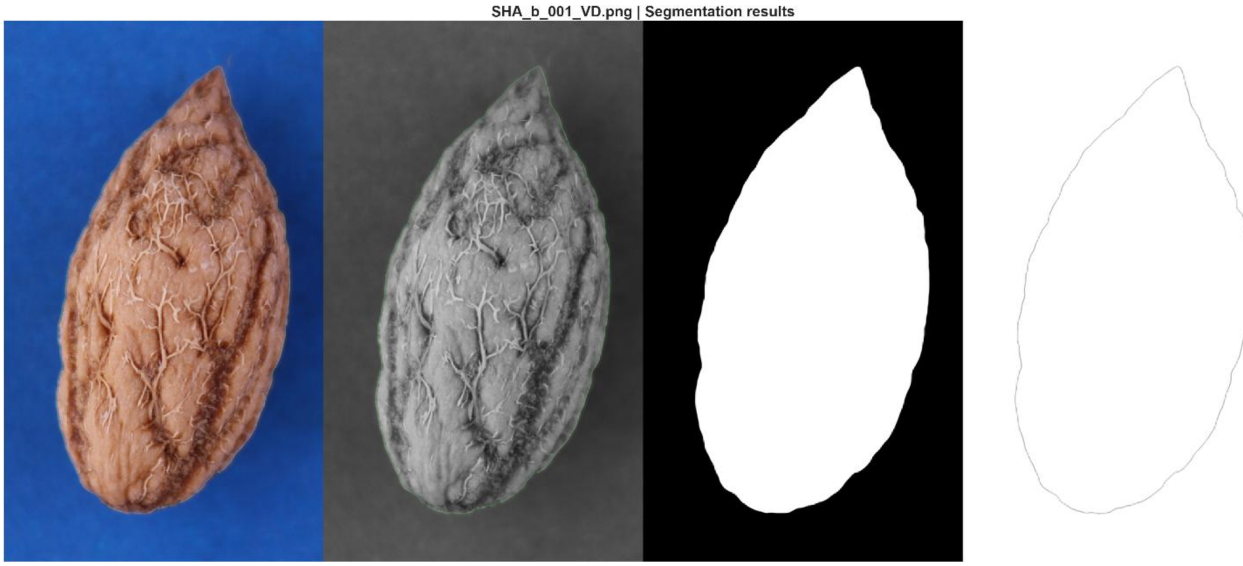
i) Mark the area to crop



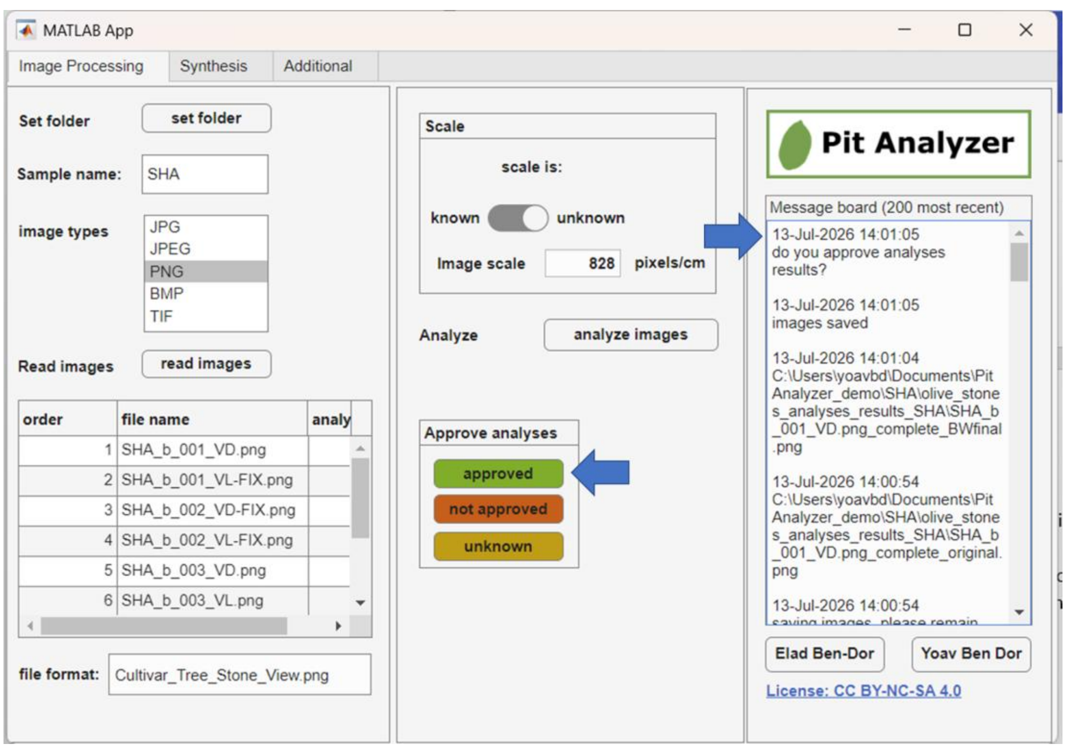
j) Press enter or click right mouse button and choose “crop image”



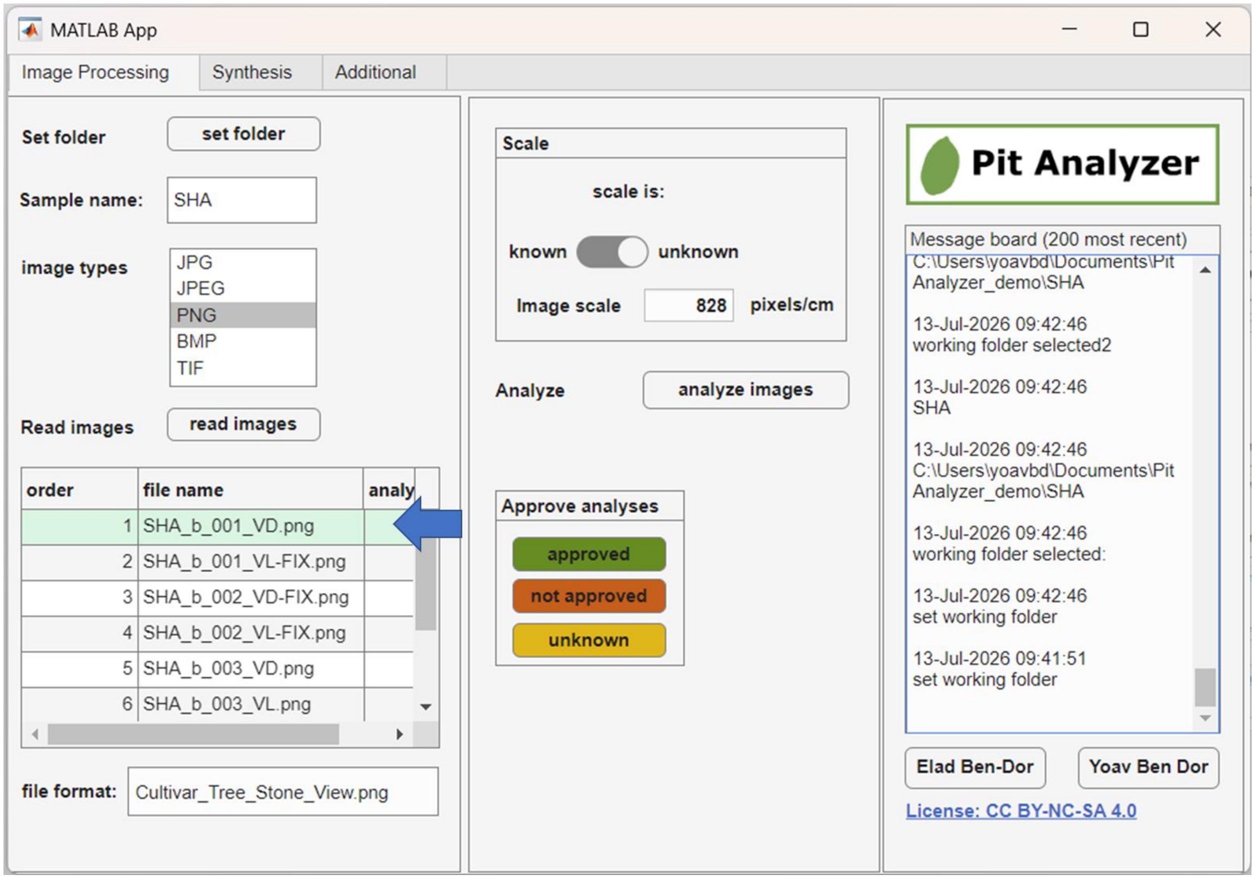
k) The program will now produce a set of images for the user to inspect, highlighting the silhouette of the pit.



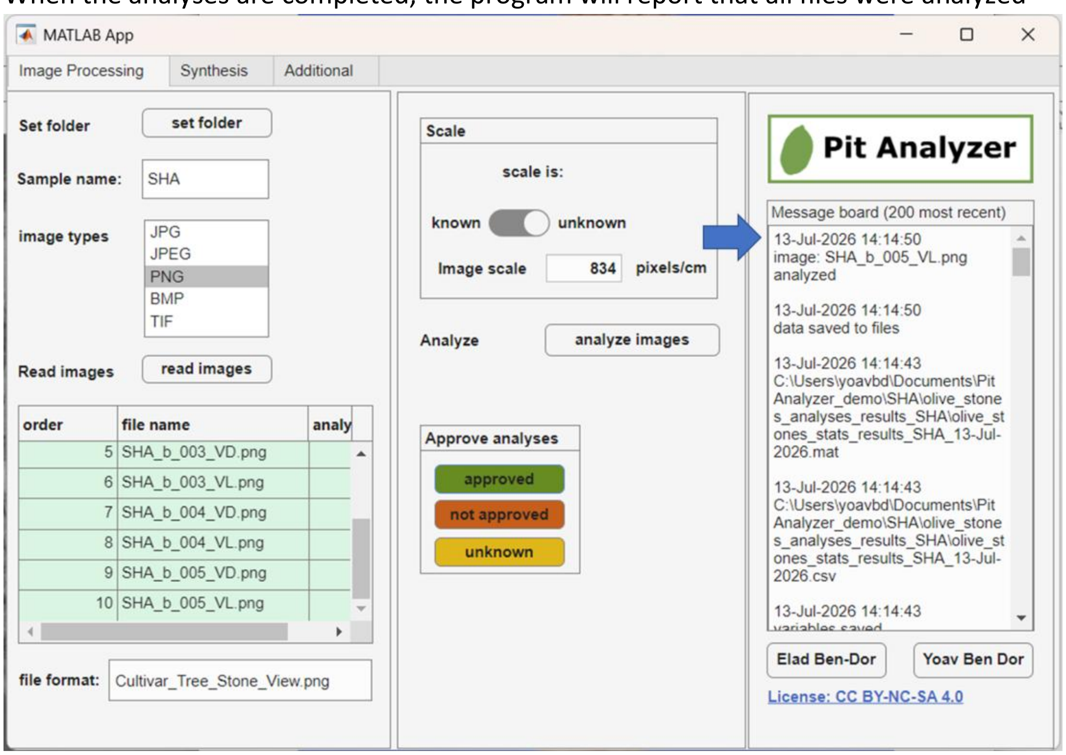
l) The user will be asked to approve or disapprove the segmentation results. This reply will be logged in the resulting csv file, and files that were not approved will be analyzed once more after replacement\correction.



m) The program will now update the files table and will automatically proceed to the next image



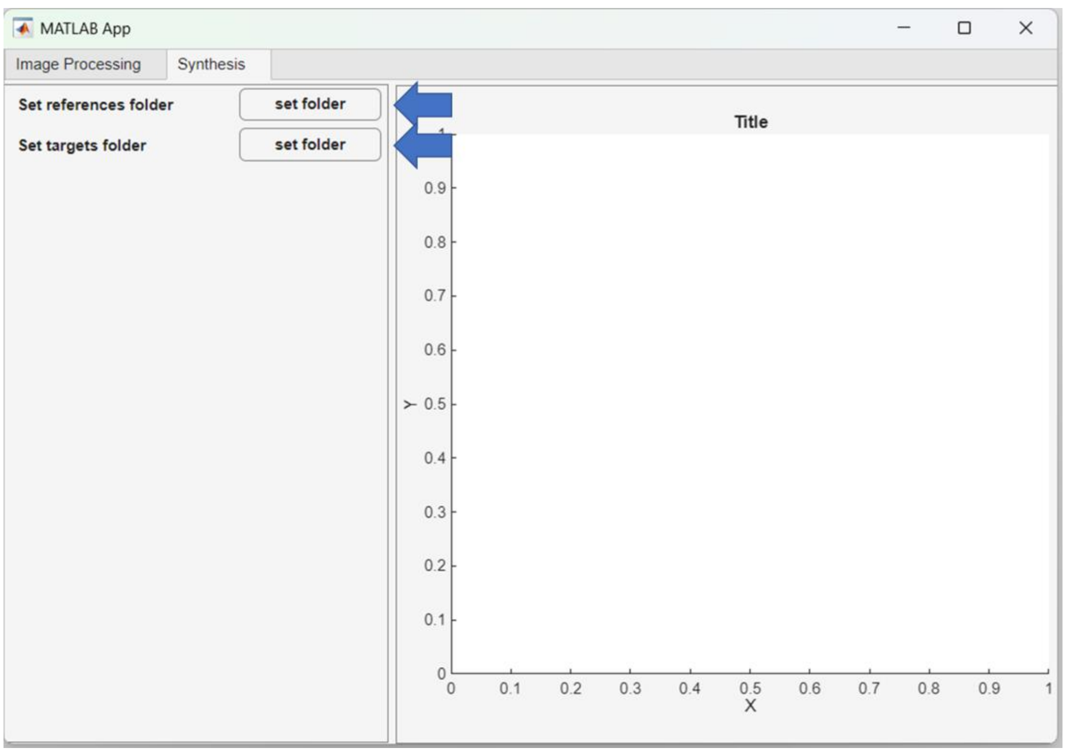
n) When the analyses are completed, the program will report that all files were analyzed



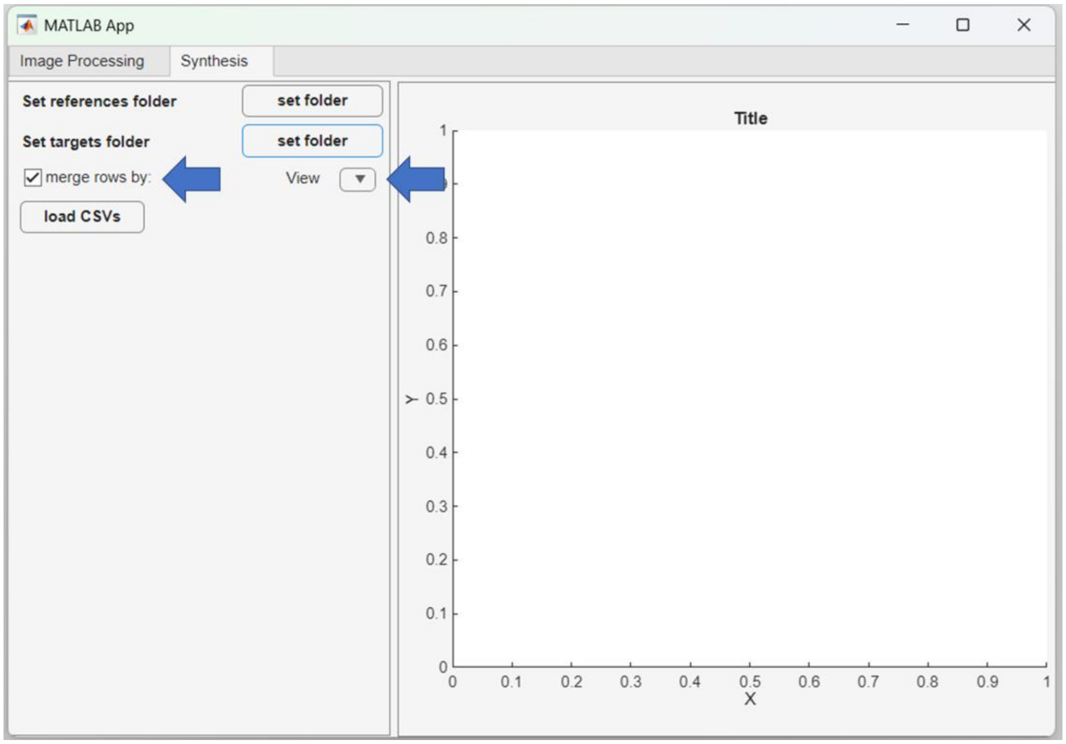
o) Tip: in order to ensure a uniform scale use a fixed camera with a fixed zoom. Make sure to place the items at the same distance for every imaging session when taking the photos.

10. Data analyses

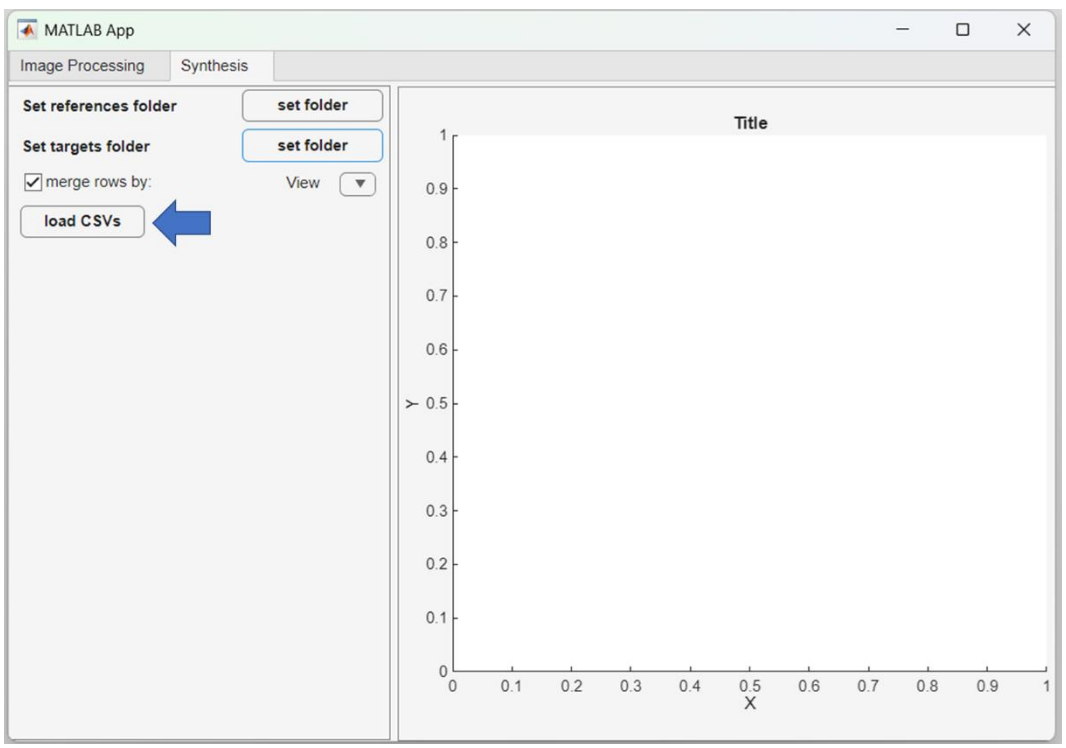
a) After completing the image processing, the user should collect all the resulting csv files into a folder for further analyses
b) The user may analyze the data using any data processing software, or using the built-in analyses tools in PitAnalyzer
c) The csv files should be saved in two distinct folders labelled as “references” and “targets”
d) This will enable specific symbology for the comparison of the results, and would also make sure the PCA is calculated using the references, and that the targets are projected using the same transformation
e) In the case that you are not comparing materials of unknown origin etc., use only the “set references folder”



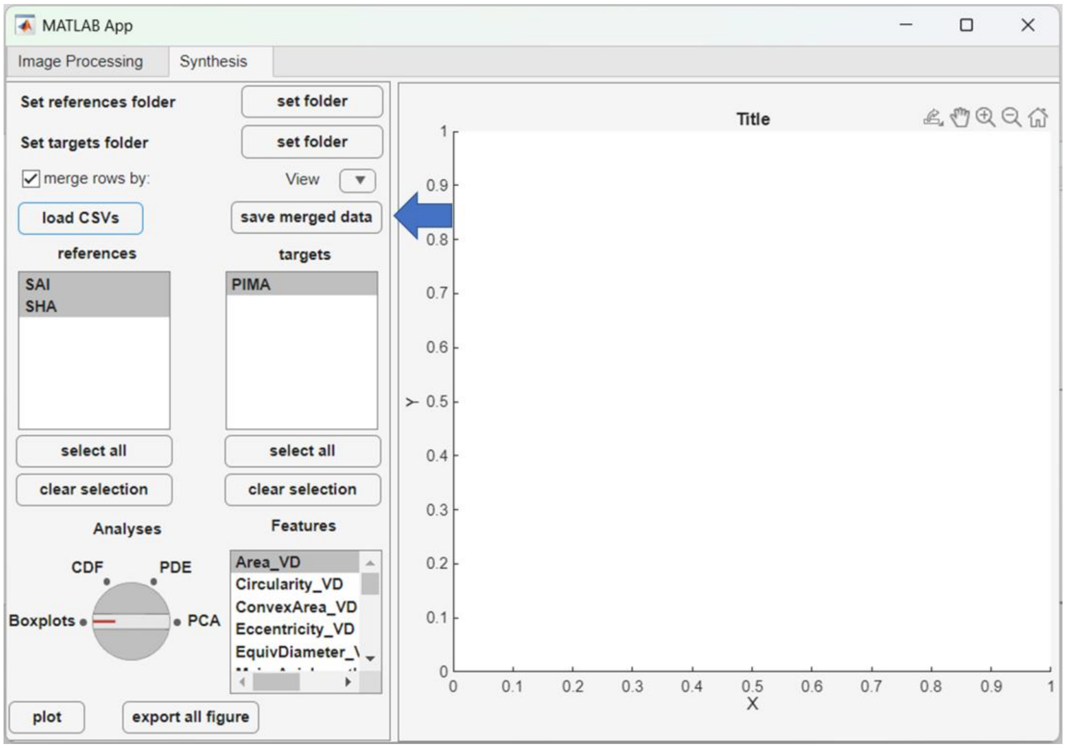
f) Define if the tables should be organized based on a specific column. By default, this should be done based on the “View” column to merge the data for the photos taken for the ventral and dorsal photos
g) If needed, change the variable to be used for reorganizing the tables



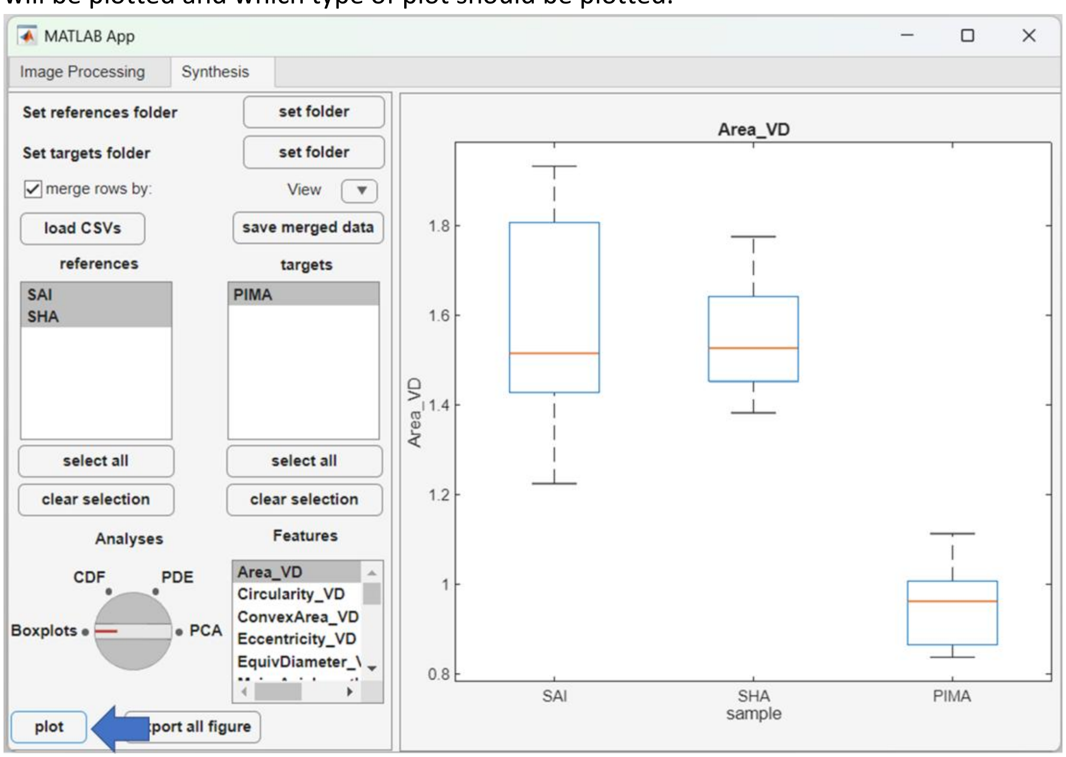
h) Click the “load CSVs” button to load the files inside the folders



i) The organized data can now saved using the “save merged data”

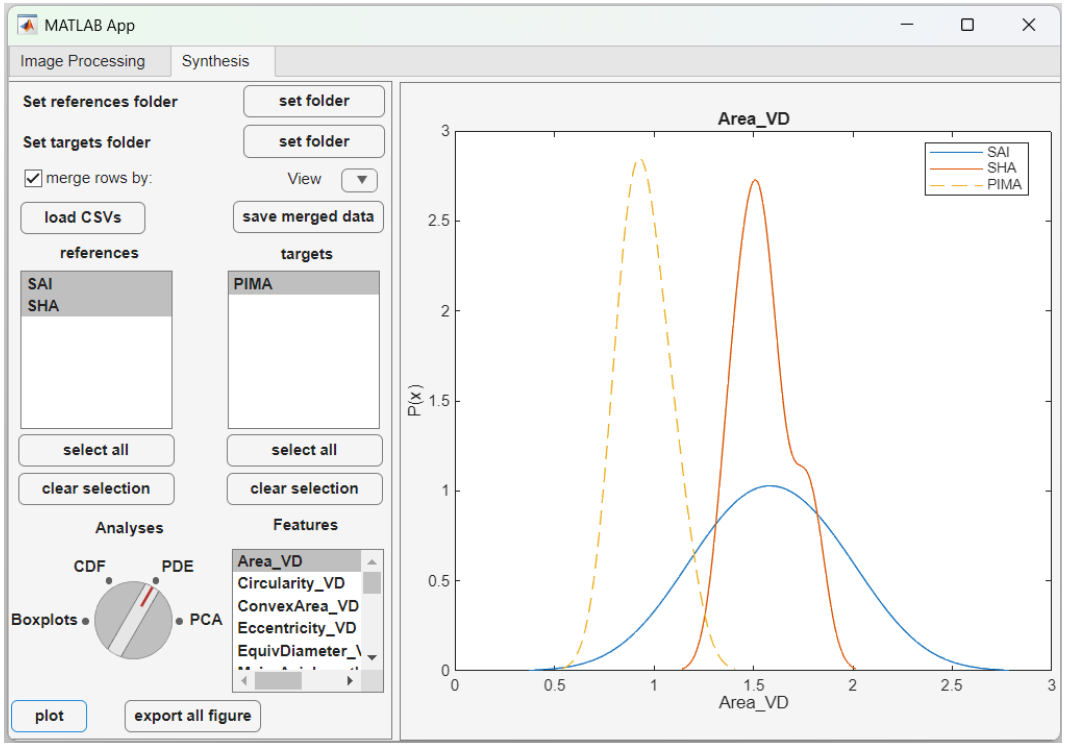


j) The program will update the menu for data analyses. The user can now choose which samples will be plotted and which type of plot should be plotted.

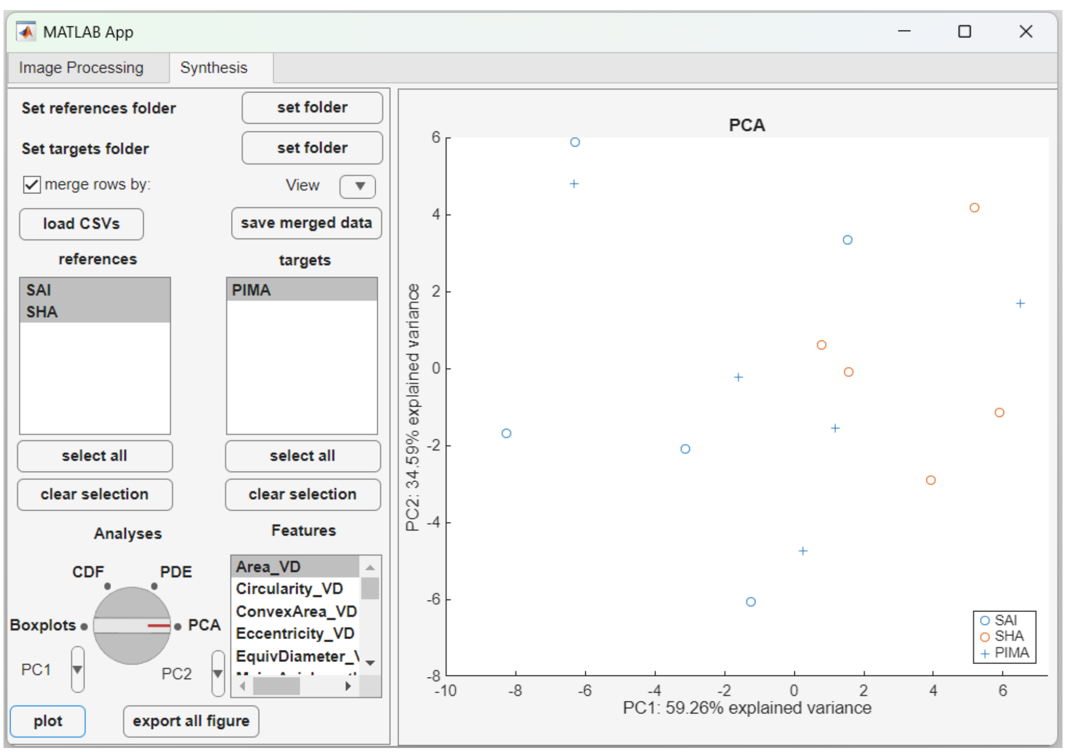


k) Different variables can be plotted in various forms such as boxplots, CDF, PDE and PCA

l) When plotting line plots (e.g., cdf and pde), targets will be plotted with a dashed line

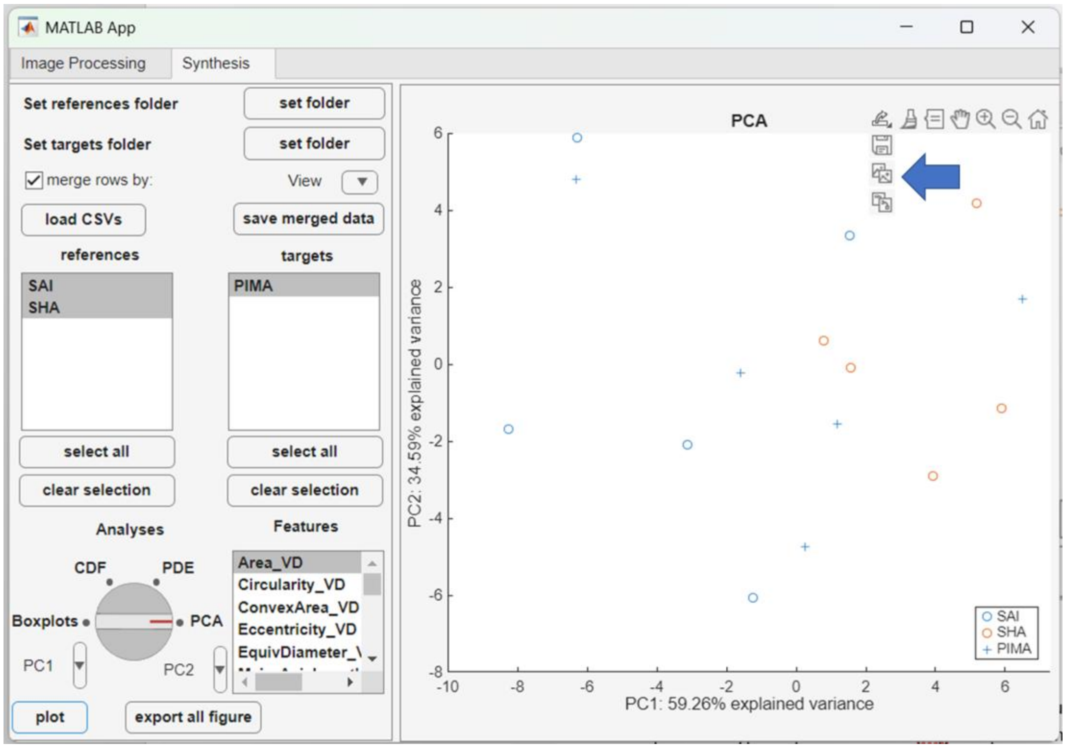


m) When using the PCA option, different symbols will be used to distinguish references from targets



n) The user can toggle the different PCs to be plotted

o) Each plot can be exported using “save as” button on the top of the figure



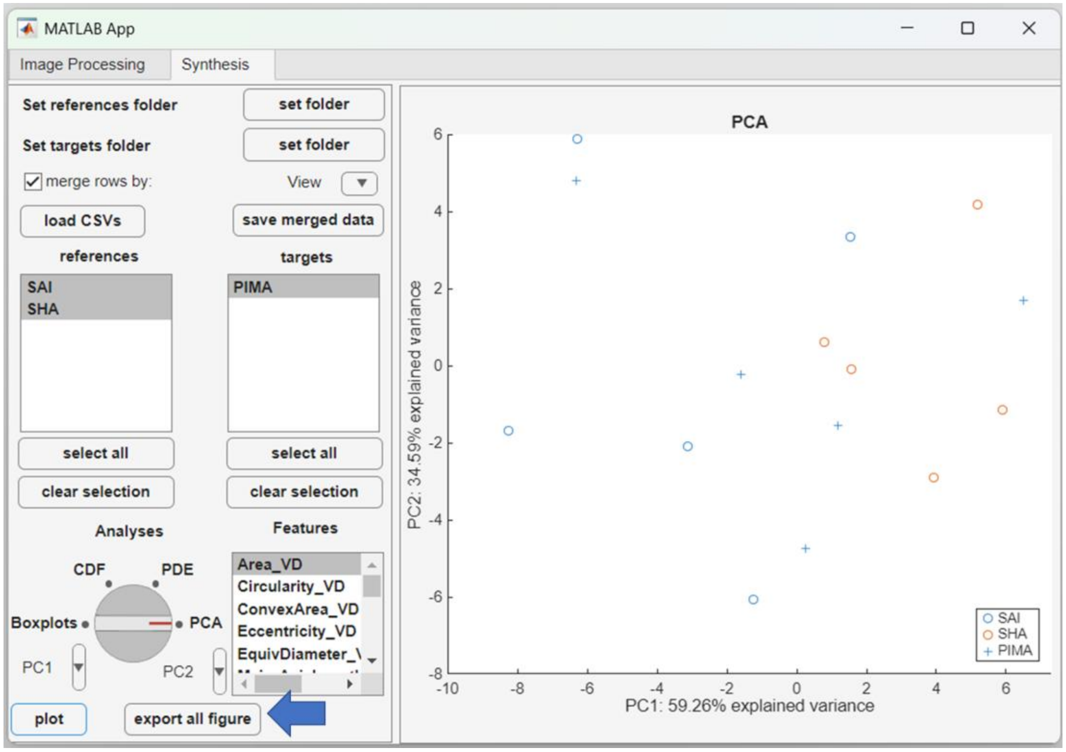
p) Tip: verify all files have the same column headers prior to running the analyses

11. Export all figures

a) Clicking the "export all figures button" will produce and export the entire set of figures of all features based on the selected samples and type of plot into both figure (PNG) and vector (PDF) formats

b) Please keep in mind that using this option could take a while, especially for large datasets

c) Tip: the user can stop the export all process using the "break" button that will appear once the "export all figures" button is pressed



d) After pressing the “export all figures”, the user can stop the operation by pressing the “break” button